

Elementary combinatorics and the binomial theorem

Sum rule · product rule · permutations · combinations · binomial
coefficients

Today's goal: turn a counting question into a clear plan, then use a rule that
counts every allowed outcome once.

Session	Date
7 of 15	Tue 15 Sep 2026

Where session 7 sits

1 · Course overview and school-maths refresher

2 · Sets and set operations

3 · Functions

4 · Function families

5 · Theorem proving

6 · Mathematical induction

7 · Elementary combinatorics and binomial theorem

8 · Midterm

9 · GCD and divisibility

10 · Modular arithmetic

11 · Vectors

12 · Polynomials

13 · Complex numbers

14 · Course recap and exam preparation

15 · Final exam

TODAY Count separate cases, count step by step, arrange objects, select groups, and expand powers such as $(a + b)^n$.

Start by describing what you count

BEFORE YOU CALCULATE, ANSWER THREE QUESTIONS

1. **What is one outcome?** A code, a seating order, a committee, or a hand of cards?
2. **Does order matter?** Is AB different from BA?
3. **Is repetition allowed?** May an object be used more than once?

The point

The same objects can give different answers when the outcomes are different.

A small example: choose or arrange?

From the three students A, B, and C:

QUESTION	OUTCOMES	COUNT
Choose a two-person committee.	$\{A, B\}, \{A, C\}, \{B, C\}$	3
Choose a president and a secretary.	AB, AC, BA, BC, CA, CB	6

THE DIFFERENCE For a committee, $\{A, B\}$ and $\{B, A\}$ describe the same group. For two different roles, AB and BA are different assignments.

Sum rule: count one disjoint case or another

DEFINITION Split the outcomes into two cases that do not overlap. If one case has m outcomes and the other has n , then the total is

$$m + n.$$

Cases that do not overlap are called **disjoint**.

EXAMPLE A student chooses one elective. There are 3 programming electives and 2 design electives, and no course is in both lists.

$$3 + 2 = 5$$

possible choices.

Product rule: count a sequence of choices

DEFINITION If a process has two stages:

- m choices for the first stage; and
- for **each** first choice, n choices for the second stage,

then there are

$$m \cdot n$$

complete outcomes.

EXAMPLE With 3 shirts and 4 pairs of trousers, there are

$$3 \cdot 4 = 12$$

outfits.

Build a code carefully

ACTIVITY 1 PAIRS

8 MIN

A code has two uppercase English letters, then three digits.

1. How many codes are possible if letters and digits may repeat?
2. How many codes are possible if no letter repeats and no digit repeats?

Assume there are **26** letters and **10** digits. The first digit position may be **0**.

Count the code positions

RULE	NUMBER OF CODES	REASON
Repetition allowed	$26^2 \cdot 10^3$	Each letter position has 26 choices; each digit position has 10 .
No repetition	$26 \cdot 25 \cdot 10 \cdot 9 \cdot 8$	The letters have 26 , then 25 , choices. The digits have 10 , then 9 , then 8 , choices.

METHOD Each code comes from one sequence of five choices. This is why the product rule works.

Factorials count arrangements

DEFINITION For a positive integer n ,

$$n! = n(n - 1)(n - 2) \cdots 2 \cdot 1.$$

It counts the ways to arrange all n **distinct** objects in a row.

EXAMPLE Five distinct books can be arranged in

$$5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$$

orders.

THE EMPTY CASE By definition, $0! = 1$. There is exactly one way to arrange zero objects: do nothing.

Permutations: arrange some of the objects

DEFINITION A **permutation** is a selection in which order matters. The number of ways to choose and order r distinct objects from n distinct objects is

$$P(n, r) = n(n - 1) \cdots (n - r + 1) = \frac{n!}{(n - r)!}.$$

EXAMPLE From 6 students, assign president, secretary, and treasurer:

$$P(6, 3) = 6 \cdot 5 \cdot 4 = 120.$$

WHY The roles are different, so order matters.

Which counting model?

ACTIVITY 2 GROUPS OF THREE

10 MIN

For each question, choose the correct model: permutation, combination, or neither. Explain why.

1. From 8 distinct books, place 5 on a shelf in a row.
2. From 8 distinct books, choose 5 for a reading list.
3. From 6 students, assign president, secretary, and treasurer.

Then calculate the answers for questions 1 and 3.

Answers and the key distinction

QUESTION	MODEL	COUNT
Place 5 of 8 books on a shelf.	Permutation	$P(8, 5) = 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 = 6,720$
Choose 5 of 8 books for a list.	Combination	Order does not matter.
Assign three different roles from 6 students.	Permutation	$P(6, 3) = 120$

THE BRIDGE TO COMBINATIONS The same five books can make $5! = 120$ different shelf orders, but they make only one five-book reading list.

SECTION 2

Combinations: select without order

When order does not matter, the same group can appear in many different orders.

Next, we will remove this repeated counting and introduce the binomial coefficient.



Combinations and binomial coefficients

DEFINITION When order does not matter, the number of ways to choose r objects from n distinct objects is

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}, \quad 0 \leq r \leq n.$$

It is read as " n choose r ."

WHY DIVIDE Why divide by $r!$? In $P(n, r)$, each chosen group appears in $r!$ different orders.

EXAMPLE ABC, ACB, BAC, BCA, CAB, and CBA are the $3! = 6$ orders of the same three-person group.

A five-card hand with exactly two aces

THE MODEL A standard deck has 4 aces and 48 other cards.

A five-card hand has no order. To make a hand with exactly two aces:

1. choose 2 of the 4 aces;
2. choose 3 of the 48 other cards.

THE COUNT So the number of hands is

$$\binom{4}{2} \binom{48}{3} = 6 \cdot 17,296 = 103,776.$$

Order or no order?

ACTIVITY 3 PAIRS

10 MIN

1. A four-digit PIN uses four different digits. Zero is allowed in the first position. How many PINs are possible?
2. From **20** candidates, choose **3** student representatives. How many groups are possible?
3. From **20** candidates, choose a chairperson, a secretary, and a treasurer. How many assignments are possible?

For each answer, write the rule or formula you used.

Answers

QUESTION	COUNT	WHY
Four-digit PIN, no repeated digits	$P(10, 4) = 10 \cdot 9 \cdot 8 \cdot 7 = 5,040$	The positions have an order.
Three representatives from 20	$\binom{20}{3} = 1,140$	Only the group matters.
Three different roles from 20	$P(20, 3) = 20 \cdot 19 \cdot 18 = 6,840$	Changing roles gives a different assignment.

THE FACTOR BETWEEN THEM For the same three people, there are $3! = 6$ different ways to assign the three roles.

A useful identity: choose or leave out

IDENTITY For $0 \leq k \leq n$,

$$\binom{n}{k} = \binom{n}{n-k}.$$

ALGEBRAICALLY

$$\frac{n!}{k!(n-k)!} = \frac{n!}{(n-k)!k!}.$$

BY COUNTING Choosing the k people who attend determines the $n - k$ people who do not attend. Each choice determines the other.

Double counting: all subsets of an n -element set

BY SIZE For each $k = 0, 1, \dots, n$, there are $\binom{n}{k}$ subsets with exactly k elements. Add these numbers:

$$\binom{n}{0} + \binom{n}{1} + \dots + \binom{n}{n}.$$

BY DECISIONS For each of the n elements, decide "in" or "out." This gives 2^n subsets.

THEREFORE Both methods count the set of all subsets of an n -element set, so

$$\sum_{k=0}^n \binom{n}{k} = 2^n.$$

■

SECTION 3

Pascal's triangle and the binomial theorem

The same binomial coefficients that count groups also appear when we expand $(a + b)^n$.

We will first see the coefficients in Pascal's triangle, then explain why they occur.



Pascal's triangle

DEFINITION Row n contains the binomial coefficients $\binom{n}{0}, \binom{n}{1}, \dots, \binom{n}{n}$.

PASCAL'S IDENTITY Every entry that is not at an edge is the sum of the two entries directly above it:

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}, \quad 1 \leq k \leq n-1.$$

n	COEFFICIENTS					
0	1					
1	1	1				
2	1	2	1			
3	1	3	3	1		
4	1	4	6	4	1	
5	1	5	10	10	5	1

The binomial theorem

THEOREM For every non-negative integer n (that is, $n = 0, 1, 2, \dots$) and real numbers a, b ,

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k.$$

WRITTEN OUT The expansion starts and ends as follows:

$$(a + b)^n = \binom{n}{0} a^n + \binom{n}{1} a^{n-1} b + \dots + \binom{n}{n} b^n.$$

READ THE EXPONENTS The exponent of a goes down from n to 0 . The exponent of b goes up from 0 to n .

Worked expansion: $(x + 2)^5$

PASCAL'S ROWPascal's row for $n = 5$ is $1, 5, 10, 10, 5, 1.$ **SO**

$$\begin{aligned}(x + 2)^5 &= x^5 + 5x^4(2) + 10x^3(2^2) \\ &\quad + 10x^2(2^3) + 5x(2^4) + 2^5 \\ &= x^5 + 10x^4 + 40x^3 + 80x^2 + 80x + 32.\end{aligned}$$

Why is the coefficient $\binom{n}{k}$?

PROOF There are n factors, each equal to $(a + b)$:

$$(a + b)^n = (a + b)(a + b) \cdots (a + b).$$

To make the term $a^{n-k}b^k$:

- choose b from exactly k factors;
- choose a from the other $n - k$ factors.

There are $\binom{n}{k}$ ways to choose those k factors. So $\binom{n}{k}$ is the coefficient of $a^{n-k}b^k$. ■

EXAMPLE In $(a + b)^3$, the term ab^2 comes from choosing b from exactly two factors. The possible choices are factors 12, 13, and 23, so its coefficient is $3 = \binom{3}{2}$.

Expand with the theorem

ACTIVITY 4 WORK INDEPENDENTLY

8 MIN

Use the binomial theorem to expand

$$(2x - 3)^4.$$

Checklist:

1. Write the row of Pascal's triangle for $n = 4$.
2. Keep the negative sign with -3 .
3. Write the whole term before you simplify it.

Walkthrough

THE EXPANSION Using the coefficients 1, 4, 6, 4, 1,

$$\begin{aligned}(2x - 3)^4 &= (2x)^4 + 4(2x)^3(-3) + 6(2x)^2(-3)^2 \\ &\quad + 4(2x)(-3)^3 + (-3)^4 \\ &= 16x^4 - 96x^3 + 216x^2 - 216x + 81.\end{aligned}$$

THE SIGNS The coefficients of x^3 and x are negative because they contain an odd power of -3 .

Pigeonhole principle

DEFINITION If more than m objects are put into m boxes, at least one box has at least two objects.

EXAMPLE Among 13 people, at least two were born in the same month.

- objects: the 13 people;
- boxes: the 12 birth months;
- since $13 > 12$, at least one month has at least two people.

DO NOT OVERSTATE IT This shows a shared month, not necessarily a shared birthday.

Explain your method

1 A café offers 4 hot drinks and 3 cold drinks. A customer chooses one drink. How many choices are there, and which rule applies?

2 From 9 distinct students, how many three-person committees are possible?

3 Find the coefficient of x^3 in $(x + 2)^5$.

4 Explain in one sentence why $\binom{8}{3}$ is not the answer to "arrange 3 of 8 books on a shelf."

RULE Write one sentence of reasoning for each answer.

What today was about

- Use the **sum rule** for separate alternatives and the **product rule** for steps in a process.
- Use a **permutation** when order matters; use a **combination** when it does not.
- $\binom{n}{k}$ counts selections of k objects from n distinct objects.
- Pascal's triangle gives binomial coefficients, and the **binomial theorem** expands $(a + b)^n$.
- Counting one collection in two ways can prove an identity.

sum rule

product rule

permutation

combination

factorial

binomial coefficient

Pascal's triangle

double counting

pigeonhole principle

Problem Set 7

Ten counting problems. For each answer, name the rule you use. Then explain why the cases do not overlap or why order matters.

Next session: **Midterm** — Sessions 1–7.